# Entomology

***Lecturer: Asmaa Khalis***
***Hawler Medical University***
***College of Medicine***
***Dept: Microbiology& Physiology***
***LG9  1 hour***

# Entomology

❖**Entomology** is a science that deals with the study of **arthropods** in general, and this science like zoology, biology, parasitology and micro-biology.

❖ **Medical entomology:**

❖Is a branch of entomology which deals with **arthropods** which affect the health of human and animals, or is a medical science directly concerned with vectors that affect human and animal health.

❖**Phylum Arthropoda** are the most numerous and widely distributed of all animal groups. Their medical importance lies in their ability to cause morbidity and mortality, They may be found in every part of the world and in every type of environment. live in close association with humans; or adapt as human parasites.

# Arthropods

**Arthropods:** "**Arthro**" means jointed and "**Poda**" means legs. Arthropods are invertebrate animals with jointed appendages((legs,antennae and mouthparts).

**Characteristic features:**

1. Hard outer covering or exoskeleton (**cuticle**) made of chitin which helps for the protection of the muscles.
2. These organisms are characterized by having of jointed appendages.
3. Body cavity called **haemocoelom**, which contains haemolymph **(blood & lymph**).
4. Some have a **cephalothorax** (head and thorax are fused).
5. All arthropods **molt** or **ecdysis** (shed exoskeleton). Were by the cuticle is shed in order to growing tissues. **(shedding off an outer layer or covering and the formation of its replacement)**.

## The effect of arthropods to humans:

1. Blood sucking (**mosquitoes**).

2. Sting and inoculation of poison (**scorpion)**.

3. Itching (Flea, ***Pulex irritans***).

4. Allergic reaction. **(Itch mite,*Sarcoptes scabiei*)**

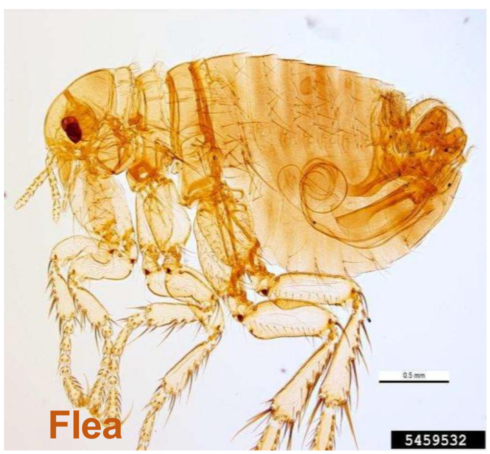
# Medical important arthropods:

There are three medically important classes of arthropods:

**1. Class:** Insect (**mosquitoes, flies, fleas, lice**).

**2. Class:** Crustacea (**craps or crayfish, Cyclopes**).

**3. Class:** Arachnida (**spiders, mites, ticks, scorpions**).




## Morphology:

Body divided into head, thorax and abdomen.

- ❖ **Head** (**feeding, sensory**), a pair of antennae, a pair of compound eyes, some with simple eyes.
- ❖ **Thorax** (**locomotion, respiration**), Only three segments, **prothorax, mesothorax** and **metathorax.**
  - ➢ Three pairs of legs, each pair attached to the thoracic segment.
  - ➢ A pair of wings (if present) attached $2^{nd}$ & $3^{rd}$ thoracic segments.
  - ➢ Breathe by spiracles / tracheae.
  - ➢ Body cavity hemocoelomic.
- ❖ **Abdomen** (**ingestion, reproduction**). Genitalia for identification.

- Life cycle shows metamorphosis either:

  **Complete metamorphosis**: egg – larvae – pupa – adult.

  (**flea, flies, mosquitoes**)

Or. **Incomplete metamorphosis**: egg – nymph – adult.

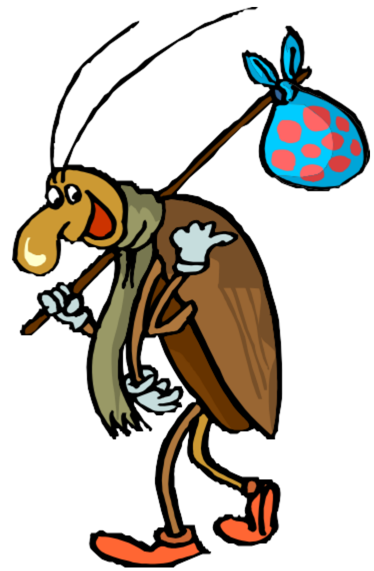
  (**lice, cockroach, Sarcoptes**).

**Nymph**: Immature stage, resembles the adult, but smaller, wingless, and with incomplete reproductive system.

## Types of mouth part:

Three types of mouth parts:

1. **Chewing** (beetles, cockroaches).
2. **Piercing-sucking** (mosquitoes, bugs,Triatoma).
3. **Lapping and sponging** ( houseflies, **Musca domestica**).



## Vector:

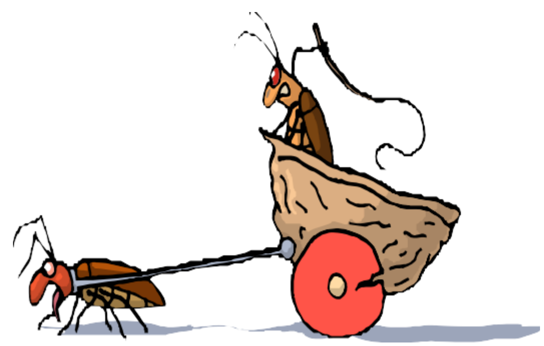
- The most important effect insects have directly on human disease is a **vector**.
- Vector arthropods may be **ticks** or **lice**, but most commonly are **flies**.

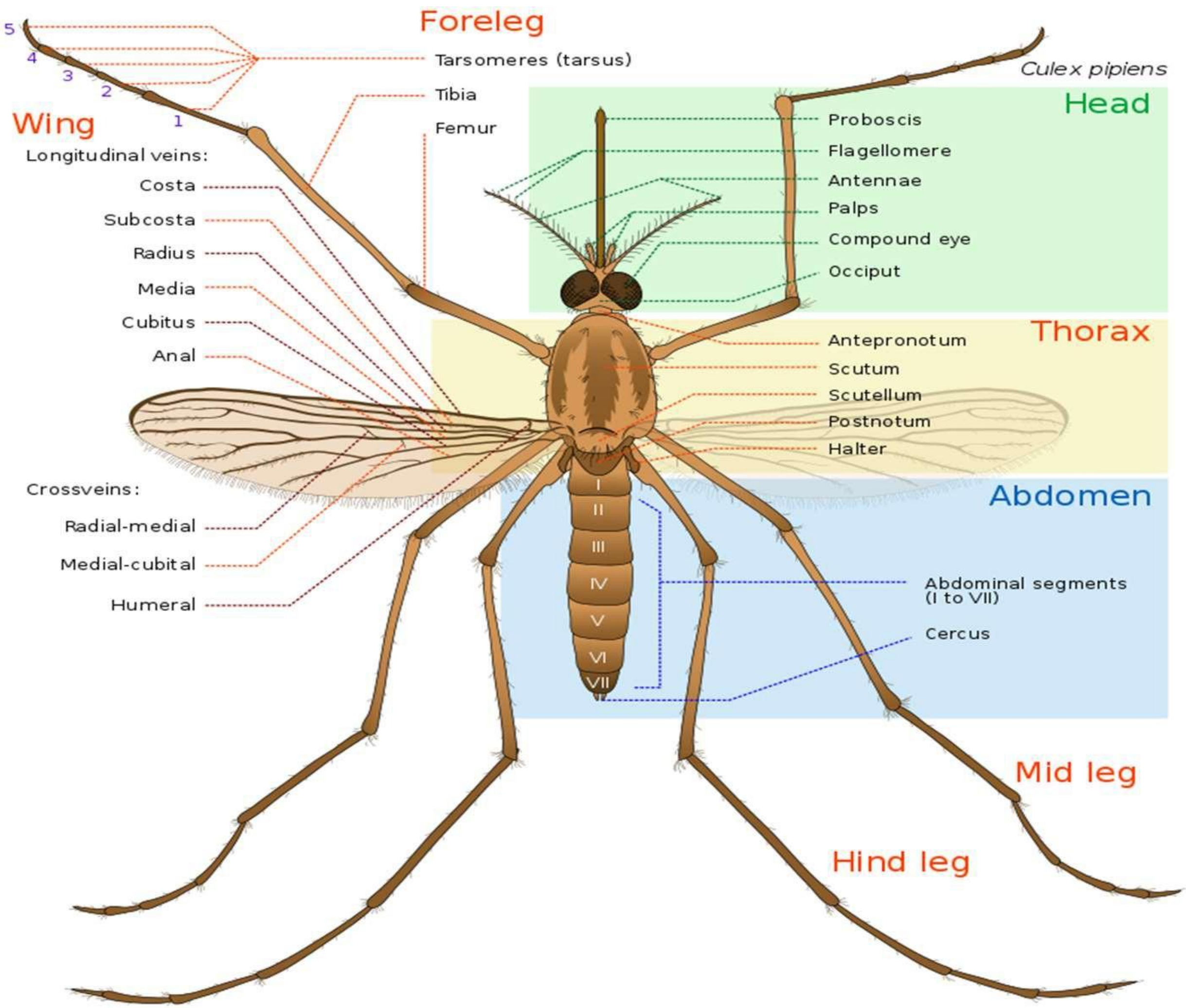
1. **Biological Vector** – It means that certain stages in the life cycle of parasite takes place in the body of the insect, e.g. **Anopheles** mosquitoes in malaria.

2. **Mechanical Vector** – Insect that carries pathogen by mouthparts or other parts of the body, but does not allow for multiplication (transmission of dysentery by **housefly**).

# Mosquitoes

- Mosquitoes have a slender segmented body and with a very long slender legs.
- The head contains a large pair compound eyes, a pair of antennae, and a single long proboscis for feeding.
- Three pairs of legs, and elongated mouthparts adapted to piercing & sucking blood.
- Females only feed on blood, male only drink sugary fluids, nectar.
- There are different species of mosquitoes (**Anopheles**, **Aedes, Culex**).

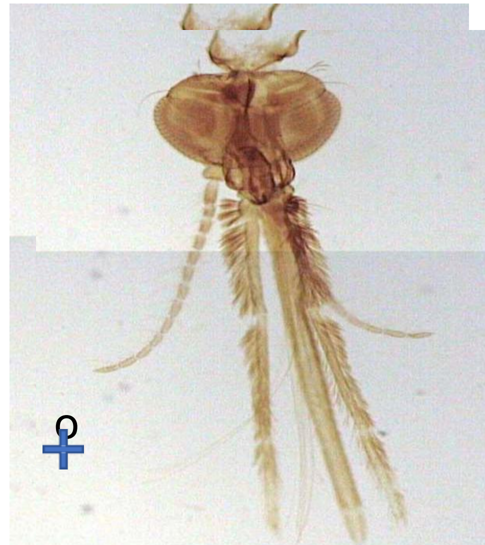


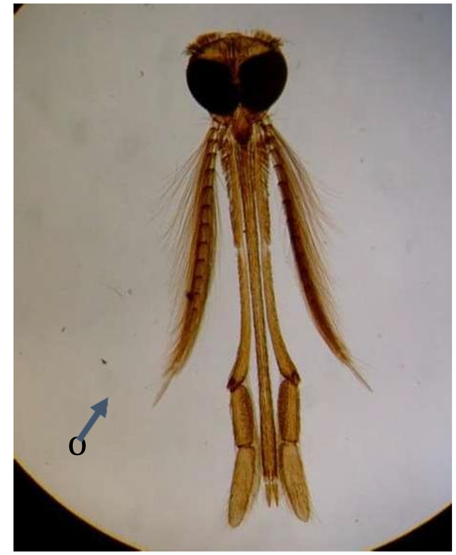


Foreleg
5
4
3
2
1
Tarsomeres (tarsus)
Tibia
Femur
Culex pipiens
Wing
Longitudinal veins:
Costa
Subcosta
Radius
Media
Cubitus
Anal
Crossveins:
Radial-medial
Medial-cubital
Humeral
Head
Proboscis
Flagellomere
Antennae
Palps
Compound eye
Occiput
Thorax
Antepronotum
Scutum
Scutellum
Postnotum
Halter
Abdomen
I
II
III
IV
V
VI
VII
Abdominal segments (I to VII)
Cercus
Mid leg
Hind leg

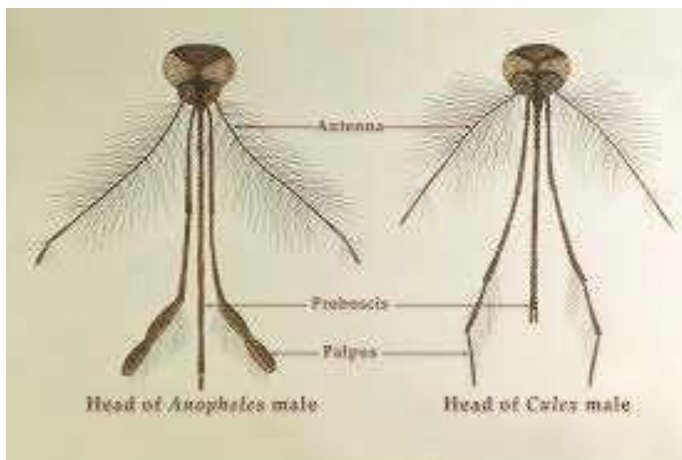


*Culex*

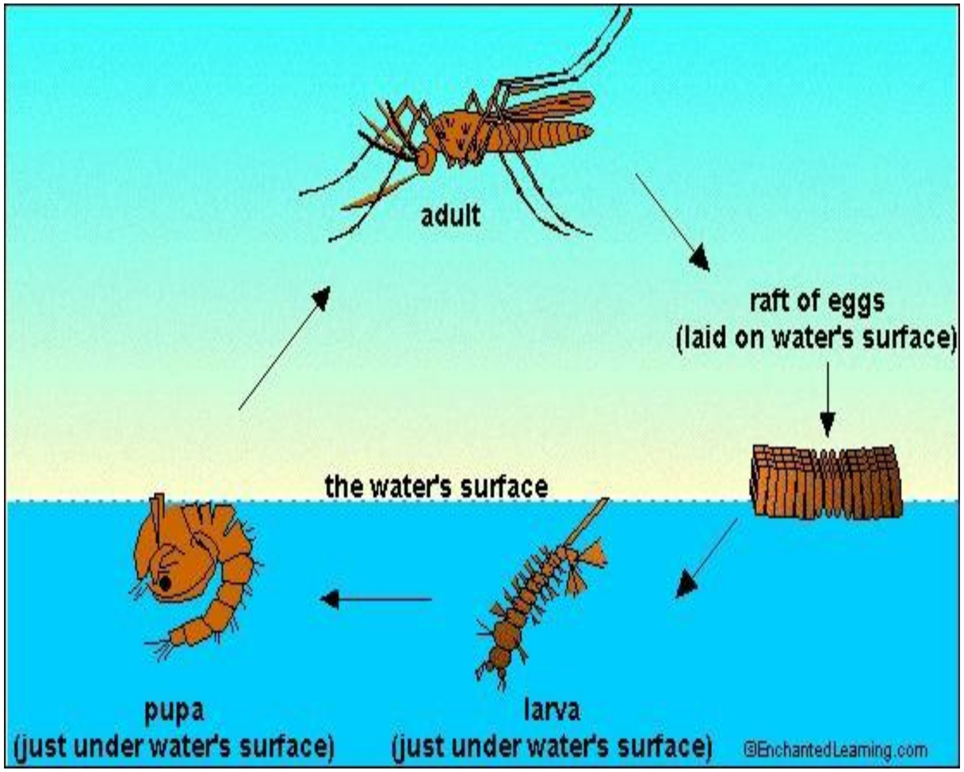



*Anopheles*

| | Anpheles. | Culex. | Aedes |
|---|---|---|---|
| **Egg** | boat-like with float (surface) | conical in raft (surface ) | fusiform single (bottom) |
| **Larva** | 2 spiracles | siphon | siphon |
| **Adult** (resting position) | body forms an angle to resting place | body is parallel to resting place | |



**Diseases caused by mosquitoes:**

1 **Malaria** is a serious disease and sometimes fatal disease caused by protozoan parasites of the genus ***Plasmodium*** parasites and is spread to humans through the bite of infected female mosquitoes (***Anopheles***).

- ❖ People who get malaria are typically very sick with high fevers, shaking chills & flu.
- ❖ ***Plasmodium*** five species:
- ❖ ***Plasmodium falciparum***, ***P. vivax, P. ovale, malariae, P. knowlesi.***
- ❖ ***Plasmodium falciparum*** is responsible for the majority of malaria deaths globally and it is a fatal disease. it is the most prevalent species in sub-Saharan Africa.

- People who get malaria are typically very sick with high fevers, shaking chills, and flu-like illness.
- ***Plasmodium*** five species:
- ***Plasmodium falciparum***, ***P. vivax, P. ovale, malariae, P. knowlesi.***
- ***Plasmodium falciparum*** is responsible for the majority of malaria deaths globally and it is a fatal disease.it is the most prevalent species in sub-Saharan Africa.

**2- Filariasis:** is a human disease caused by parasitic nematode worms known as filarial worms.

- Transmitted by a specific mosquitoes (***Culex***, ***Anopheles*** & ***Aedes***).

**A- Lymphatic filariasis:** 3 types, ***Wuchereria bancrofti, Brugia malayi*** & ***B.timori.***

- These worms damage the **lymphatic system,** including the lymph nodes.
- Blockage of lymphatic circulation leads to edema.
- ***Wuchereria bancrofti*** is being the most common worm which caused **elephantiasis.**
- **Elephantiasis:** accumulation of lymph, fibrosis and thickening of skin, swelling in the arms, legs, breasts, and genitals scrotum, the skin becomes thickened, coarse and warty.

## B. Subcutaneous Filariasis:

1 **River blindness** caused by ***Onchocercecas volvulus,*** transmit by **black flies** (***Simulium***). **Onchocerciasis** is a major cause of **blindness**.

2 **Loiasis:** It is caused by ***Loa loa*** (eye worm).It is transmitted by the bite **mango fly** **(*Chrysops*)*.*** The migrating adult worms in the subcutaneous tissues are pathogenic and provoke **inflammatory** reaction.

## **Flies (*Musca domestica*)**

The common housefly, lives in close association with people all over the world and are widespread.

Body divided in to three parts,: **head, thorax** and **abdomen**.

- **Head** have two antennae, a pair of compound red eyes that contain thousands of individual lenses, giving them wider vision, mouth parts (labrum, labium, mandible, and maxilla) modified to sponging and lapping, liquefy many solid foods.
- **Thorax** is the locomotor center; bears three pairs of legs, a pair wings on the 2$^{nd}$ thoracic segment, and the third segment bears the **halteres**, which help to balance the insect during flight.

## Life Cycle

**1 House fly** undergo **complete metamorphosis:**

**Egg**, **larva**, **pupa** and **adult.**

- Sexual mature female flies lay eggs on suitable material.
- The eggs hatch into tiny, whitish, legless larvae also known as **maggots** that resemble tiny worms, the **larvae** become **pupae** and the **adult**.
- The development time from egg to adult depends on the temperature of the surrounding environment.
- The life cycle within a few weeks and live an additional few weeks or months as adults.

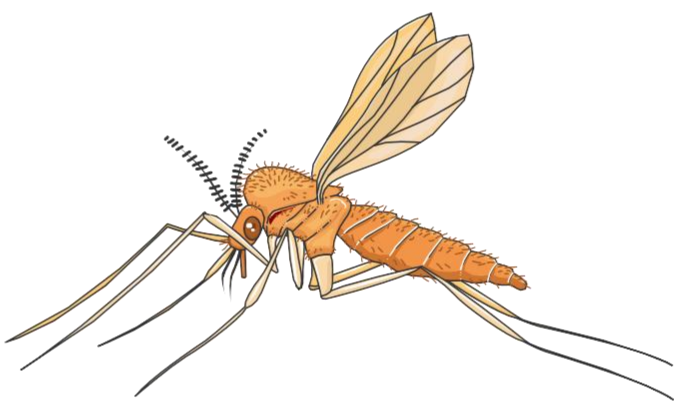
## Diseases:

**House flies** are strongly suspected of transmitting at least more than 100 different pathogens, including **salmonellosis**, **dysentery tuberculosis**, **typhoid fever**, **cholera**, **poliomyelitis**, **anthrax**, **tularemia**, **leprosy**.

- Fly can contaminate food surfaces by spreading disease picked up on their legs and mouths when feeding on trash, feces and other decaying substances.
- Flies frequently vomit on food during feeding this can lead to infection, probably the method of transmission is defecation, which often occurs on food.
- House flies also transmit a number of bacterial, viral, and protozoal diseases.

## 2- Sand fly (*Phlebotomus papatasi* ).

- There are over 600 species of sand flies divided into five genera. the genus ***Phlebotomus papatasi*** was identified as a vector of the **leishmaniasis.**
- Human are infected from being bitten by an infected female sand fly.
- Sand flies are most active in humid and warm environments at night.
- The mouthparts are well-developed with cutting teeth on elongated mandibles in the proboscis, adapted for blood-sucking in females, The abdomen is nine-segmented and tapered at the end.



**Leishmaniasis** is transmitted by protozoan parasites, small microscopic single - celled organisms from the genus ***Leishmania***.

1 **Visceral leishmaniasis (VL):** Cause by ***Leishmania donovani***. VL involving reticulo – endothelial cells of liver, spleen, lymph node and bone marrow. VL damages the internal organs and can be life - threatening if not treated, will die!

2 **Cutaneous leishmaniasis (CL):** Cause by ***Leishmania tropica*****.** CL involving endothelial cells of the capillaries of the skin.

3 **Mucocutaneous leishmaniasis (MCL):** Cause by ***Leishmania brazilliensis*****,** involving mucous membranes of the mouth and nose can lead to partial or complete total destruction of the mucosa membranes of the nose, mouth and throat cavities.

## **3- *Tse tse* fly**

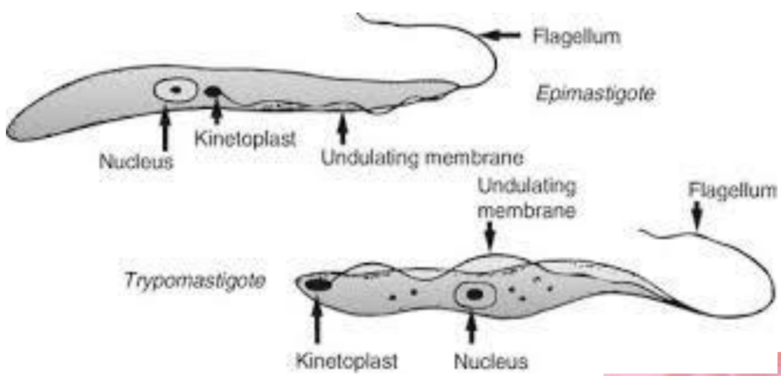
Also known as **tik-tik** flies, are large biting flies that inhabit much of tropical Africa.

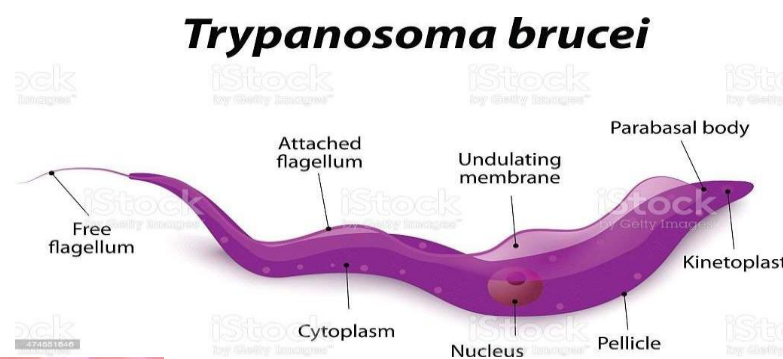
The **tsetse** are obligate parasites that live by feeding on the blood of vertebrate animals.

- **Tsetse** is a biological vector for transmitting human sleeping sickness or **trypanosomiasis.**
- **Trypanomiasis** or sleeping sickness a fatal human diseases caused by ***Trypanosoma brucei*** and chagas disease caused by ***Trypanosoma cruzi***.

- Trypanosomes are a group of unicellular parasitic flagellate protozoa.

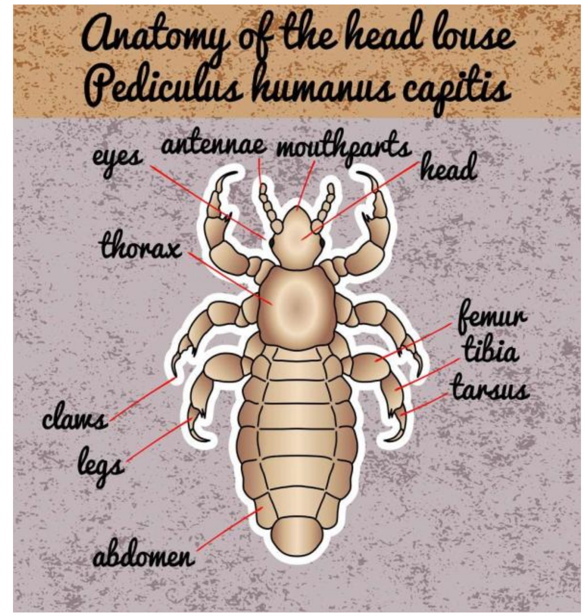
❖ The disease progresses in two stages: a lymphatic/blood stage during which the parasite is found in the lymphatic system, the "neurological phase" where the central nervous system is affected.




## Lice (*Pediculus humanus*)

- Lice are ectoparasites, and they can live in different part of the body that can be found on the head, eyebrows, and eyelashes of people.
- **Incomplete metamorphosis.**
- There are three types of lice that infest humans:
- ***Pediculus humans corporis*** ( body louse).
- ***Pediculus humans capitis*** (head louse).
- ***Phthirus pubis*** (public louse).



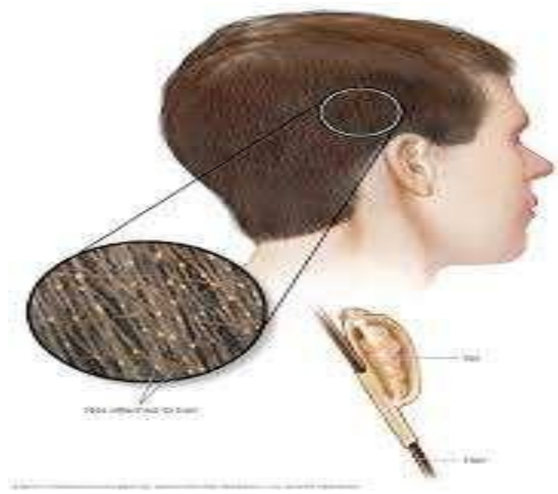

Head lice is most common among pre-school children.

Head lice are spread by direct contact with the hair of an infested person.

Spread by contact with clothing (such as hats, scarves, coats) or other personal items (such as combs, brushes, or towels).

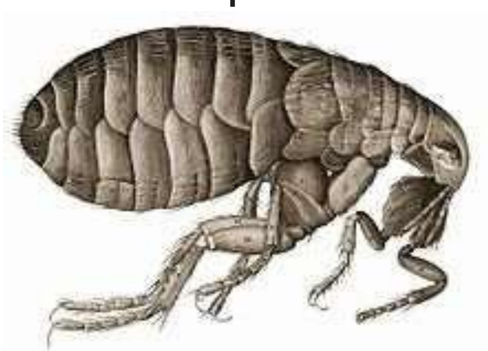
**Signs and symptoms**

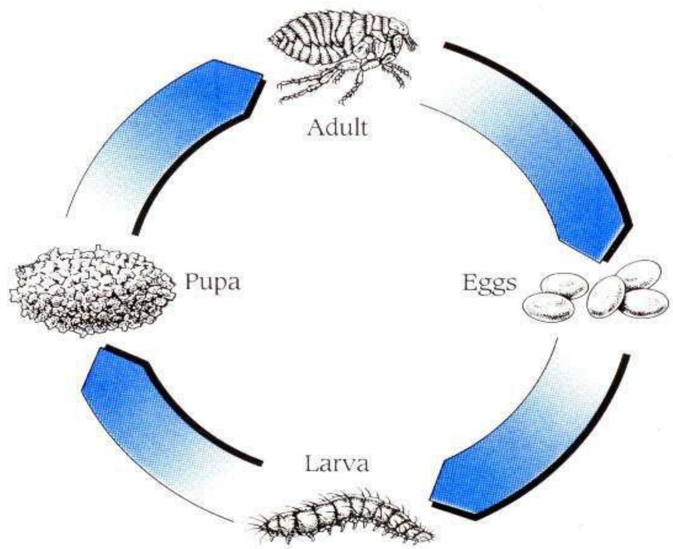
- Tickling feeling of something moving in the hair.
- Itching, caused by an allergic reaction to the bites of the head louse.
- Irritability and difficulty sleeping; head lice are most active in the dark.
- Sores on the head caused by scratching.
- Lice are also responsible for transmission of diseases such as relapsing fever and epidemic typhus.



**_Pulex irritans_ (Fleas).**

- Adult fleas are small brown, flattened, wingless insects that live as ectoparasites of mammals and birds.
- Have strong claws preventing them from being dislodged.
- Mouthparts adapted for piercing and sucking blood, live by consuming blood, or hematophagy.
- Hind legs extremely well adapted for jumping.
- With complete metamorphosis.






**Complete life cycle, fleas**

## Effects of bites

- Fleas are principally a nuisance to their hosts, causing an itching.
- Flea bites cause a slightly raised, swollen, irritating nodule to form a single puncture that can lead to an eczematous itchy skin disease called flea allergy dermatitis.

## Crustacea:

- Most crustaceans have two pairs of antennae, three pairs of chewing appendages, various numbers of pairs of legs.
- Crustaceans differ from the insects in that they have legs on their abdomen as well as on their thorax.

### Medical importance:

1) **Cyclopes** are intermediate hosts of the fish tapeworm (***Diphyllobothrium latum***) & ***Dracunculus medinensis***.
2) **Crabs** or **crayfish** are Intermediate hosts of the human lung fluke ***Paragonimus westermani***.

**Class Arachnida:** Includes 4 orders of medical importance:

1. **Scorpions.**
2. **Mite.**
3. **Spider.**
4. **Ticks**

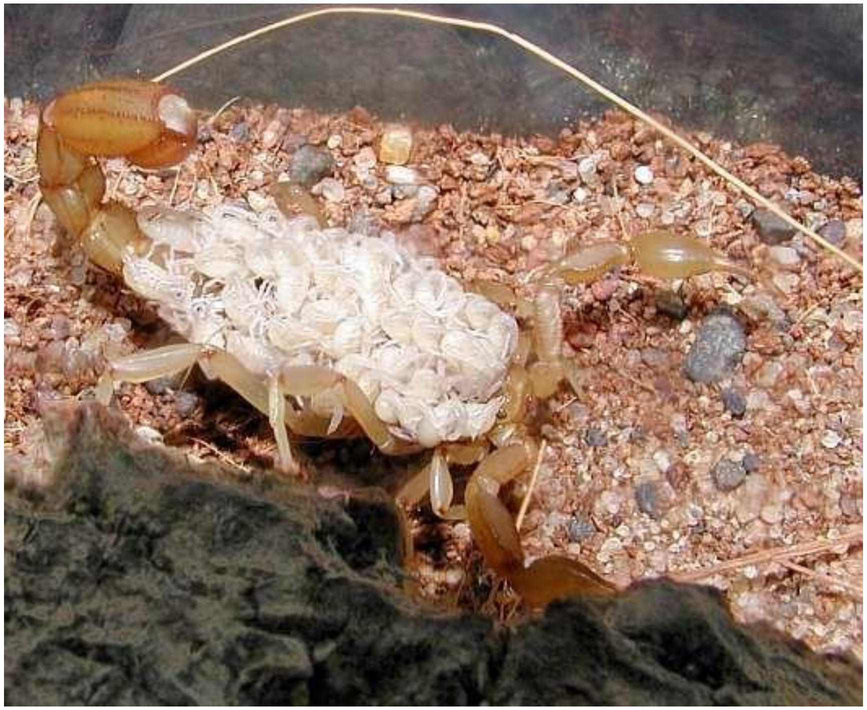


## Interesting facts about scorpions

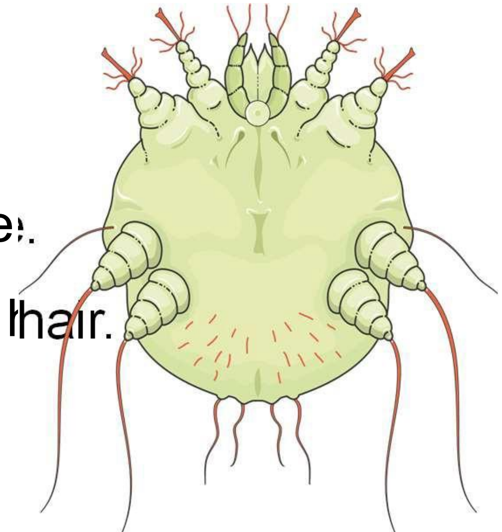
- Scorpions live everywhere, 25 types can harm/kill man.
- Scorpions have 8 legs and 2 claws.
- They also have a large stinger on the last segment of the abdomen.
- Scorpions have greatly enlarged **pedipalps**, which they hold in a forward position.
- Male and female scorpions find each other by vibration, scent, or touch. Then they dance together for half an hour or more.
- Females often eat the males when they are done with the dance.
- Venom it has.

## Scorpion Sting Symptoms

- Intense pain at the sting site.
- Mild swelling around sting site.
- Numbness in area of sting.
- Nausea or vomiting.
- Excessive salivation.



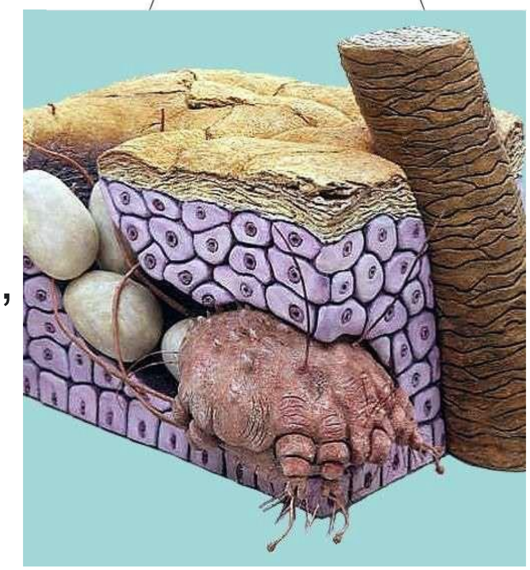
## Mite (*Sarcoptes scabiei* )

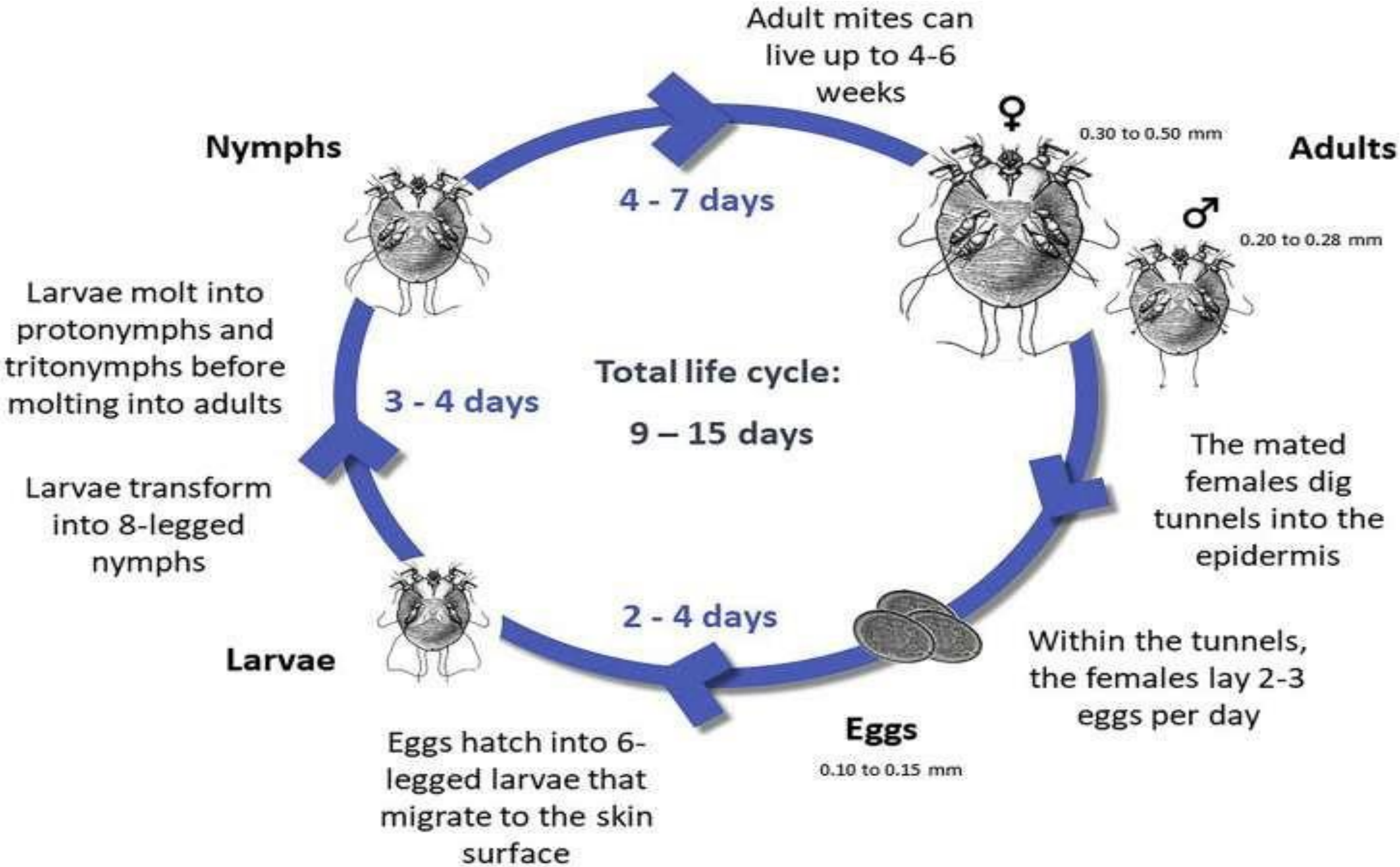


- Also called the “itch mite” itching of affected surfaces at night.
- Mites are small microscopic in size not visible with the naked eye.
- oval, gray-yellow color, dorsally convex & bears spins, bristles & hair.
- Separated sex, male smaller than the male.
- Four pairs of leg, each end with suckers.
- Mouthparts adapted for burrowing &make tunnel the skin.
- Live in **superficial epidermal** layers and **stratum corneum**.
- **Scabies** can spread quickly through **close physical** of a family,

**child care group**, **school** and **nursing home,** personal contact,

particularly holding or shaking the hands of an infested person.






**Incomplete life cycle of Mite**

## Scabies, Sign & symptoms

- Humans are bitten when they contact straw, hay, grasses, leaves, seeds or other materials harboring the mites.

  - **Allergic reaction** to to mite proteins (proteins in the **saliva, faeces** or **eggs)**.
  - Sever **dermatitis** and **irritation** which may occur on any part of the body that burrowing the thinner skin, bend of knee and elbow, between fingers, folds of the wrist, genitalia and breasts.

- Lesions are initially **erythematous redness of the skin**), then become **papular** (**skin lesion**), papules rupture and skin becomes **crusty**.

- **Alopecia**, thickening of the skin, **pruritus** (**itching**).


Thank You